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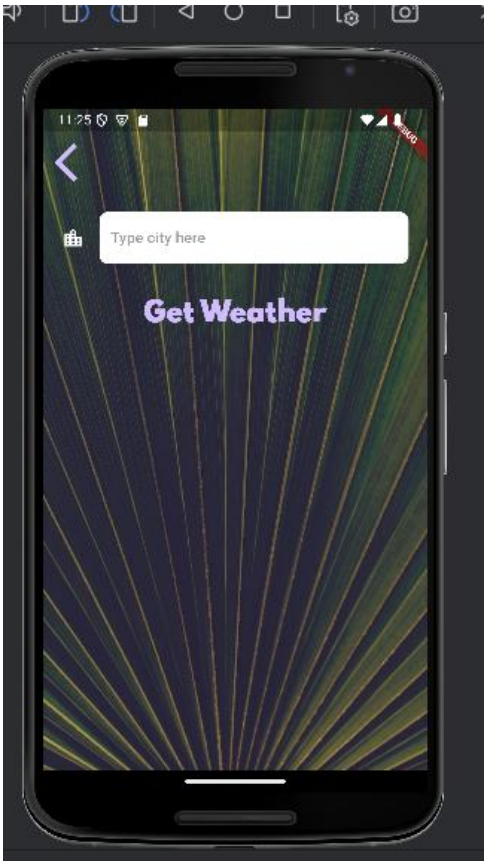
Result:



CODE:

Main.dart

```
import 'package:flutter/material.dart';
import 'screens/loading_screen.dart';

void main () => runApp(const MyApp());

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      theme: ThemeData.dark(),
      home: const LoadingScreen(),
    );
    throw UnimplementedError();
  }
```

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```
}
```

City_Screen.dart

```
import 'package:flutter/material.dart';
import 'package:test_clima_flutter/screens/loading_screen.dart';
import 'package:test_clima_flutter/services/networking.dart';
import 'package:test_clima_flutter/screens/location_screen.dart';
import 'package:test_clima_flutter/utilities/constants.dart';

import '../utilities/constant.dart';

class CityScreen extends StatefulWidget {
  const CityScreen({super.key});

  @override
  State<CityScreen> createState() => _CityScreenState();
}

class _CityScreenState extends State<CityScreen> {
  String newcity = '' ;
  @override
  Widget build(BuildContext context) {
    return Scaffold(
      body: Container(
        decoration: const BoxDecoration(
          image: DecorationImage(
            image: AssetImage('images/city_background.jpg'),
            fit: BoxFit.cover,
          ),
        ),
        constraints: const BoxConstraints.expand(),
        child: SafeArea(
          child: Column(
            children: <Widget>[
              Align(
                alignment: Alignment.topLeft,
                child: TextButton(
                  onPressed: () {
                    Navigator.pop(context);
                  },
                  child: const Icon(
                    Icons.arrow_back_ios,
                    size: 50.0,
                  ),
                ),
              ),
              Container(
                  padding: const EdgeInsets.all(20.0),
                  child:   TextField(
```

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```
                    onChanged: (value){
                      newcity = value;
                    },
                    style: const TextStyle(
                      color: Colors.black,
                    ),
                    decoration: const InputDecoration(
                        filled: true,
                        fillColor: Colors.white,
                        icon:  Icon(
                          Icons.location_city,
                          color: Colors.white,
                        ),
                        hintText: 'Type city here',
                        hintStyle: TextStyle(
                          color: Colors.grey,
                        ),
                        border: OutlineInputBorder(
                          borderRadius: BorderRadius.all(
                            Radius.circular(10.0),
                          ),
                          borderSide: BorderSide.none,
                        )
                    ),
                  )
              ),
              TextButton(
                onPressed: () {
                  Navigator.pop(context,newcity);
                },
                child: const Text(
                  'Get Weather',
                  style: kButtonTextStyle,
                ),
              ),
            ],
          ),
        ),
      ),
    );
  }
}
```

Loading_Screen.dart

```
import 'package:http/http.dart';
import 'package:flutter/material.dart';
import 'package:test_clima_flutter/services/networking.dart';
import 'package:flutter_spinkit/flutter_spinkit.dart';
import 'package:test_clima_flutter/screens/location_screen.dart';

class LoadingScreen extends StatefulWidget {
  const LoadingScreen({super.key});

  @override
```

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```
  State<LoadingScreen> createState() => _LoadingScreenState();

  void getWeatherData(String newcity) {

  }
}


class _LoadingScreenState extends State<LoadingScreen> {
  String city ='';

  @override
  void initState() {
    super.initState();
    getWeatherData();
  }

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      body: Center(
        child: SpinKitDoubleBounce(
          size: 70.0,
          color: Colors.white,
        ),
      ),
    );
  }

  void getWeatherData() async{
    Networking networking = new  Networking();
    String data = await networking.getData(city);


    Navigator.push(context, MaterialPageRoute(builder: (context){
      return LocationScreen(data);
    }));
  }
}
```

Location_Screen.dart

```
import 'package:flutter/material.dart';
import 'package:test_clima_flutter/screens/city_screen.dart';
import 'package:test_clima_flutter/utilities/constant.dart';
import 'package:test_clima_flutter/utilities/constants.dart';
import 'dart:convert';
import 'package:test_clima_flutter/services/weather.dart';
import 'package:test_clima_flutter/services/networking.dart';

class LocationScreen extends StatefulWidget {
  LocationScreen(this.data, {super.key});
  String data;

  @override
```

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```
  State<LocationScreen> createState() => _LocationScreenState();
}

class _LocationScreenState extends State<LocationScreen> {
  double temp = 0;
  int id = 0;
  String city = ' ', info = ' ', weathericon = ' ', weathermsg = ' ', update
= ' ';
  bool error = false;

  @override
  void initState() {
    super.initState();
    info = widget.data;
    updateUI();
  }

  void updateUI() {
    setState(() {
      city = jsonDecode(info)['name'];
      temp = jsonDecode(info)['main']['temp'];
      id = jsonDecode(info)['weather'][0]['id'];
      error = false;

      print(city);
      print(temp);
      print(id);
      print(info);

      WeatherModel weatherModel = new WeatherModel();
      weathericon = weatherModel.getWeatherIcon(id);
      weathermsg = weatherModel.getMessage(temp.toInt());
    });
  }

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      body: Container(
        decoration: BoxDecoration(
          image: DecorationImage(
            image: const AssetImage('images/location_background.jpg'),
            fit: BoxFit.cover,
            colorFilter: ColorFilter.mode(
                Colors.white.withOpacity(0.8), BlendMode.dstATop),
          ),
        ),
        constraints: const BoxConstraints.expand(),
        child: SafeArea(
          child: Column(
            mainAxisAlignment: MainAxisAlignment.spaceBetween,
            crossAxisAlignment: CrossAxisAlignment.stretch,
            children: <Widget>[
              Row(
                mainAxisAlignment: MainAxisAlignment.spaceBetween,
                children: <Widget>[
                  TextButton(
```

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```
                onPressed: () async {
                  Networking netting = new Networking();
                  update = await netting.getData(city);
                  setState(() {
                    info = update;
                    updateUI();
                    error = false;
                  });
                },
                child: const Icon(
                  Icons.near_me,
                  size: 50.0,
                ),
              ),
              TextButton(
                onPressed: () async {
                  String newCity;
                  newCity = await Navigator.push(context,
                      MaterialPageRoute(builder: (context) {
                        return const CityScreen();
                      }));
                  Networking networking = new Networking();

                  try {
                    update = await networking.GetCity(newCity);

                    setState(() {
                      info = update;
                      updateUI();
                      error = false;
                    });

                  }

                  catch (e) {
                    setState(() {
                      weathermsg = e.toString();
                      city = " ";
                      error = true;
                    });
                  }
                },
                child: const Icon(
                  Icons.location_city,
                  size: 50.0,
                ),
              ),
            ],
          ),
          Padding(
            padding: const EdgeInsets.only(left: 15.0),
            child: Row(
              children: <Widget>[
                if (!error)
                  Text(
                    temp.toStringAsFixed(0) + '°',
                    style: kTempTextStyle,
```

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```
                    ),
                  Text(
                    error ? 'ERROR' : weathericon,
                    style: kConditionTextStyle,
                  ),
                ],
              ),
            ),
            Padding(
              padding: const EdgeInsets.only(right: 15.0),
              child: Text(
                error ? weathermsg
                    : "$weathermsg in $city",
                textAlign: TextAlign.right,
                style: kMessageTextStyle,
              ),
            ),
          ],
        ),
      ),
    ),
  );
 }
}
```

Location.dart

```
import 'package:geolocator/geolocator.dart';
class Location{

double lon = 0, lat = 0;
  Future getLocation() async{
    try {
      LocationPermission permission = await Geolocator.requestPermission();
      Position position = await
Geolocator.getCurrentPosition(desiredAccuracy: LocationAccuracy.low);
      lon = position.longitude;
      lat = position.latitude;
    } catch (e) {
      print(e);
    }
  }
}
```

Networking.dart

```
import 'package:http/http.dart';
import 'package:test_clima_flutter/services/location.dart';
import 'package:flutter/foundation.dart';

class Networking{
```

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```
  String appID = '04c2edac80d11d65ae5040192beaa7f7';
  double temp=0;
  String data='', city='';
  int id=0;

  Future<String> getData(city) async{
    Location location =  new Location();
    await location.getLocation();
    double lat = location.lat;
    double lon = location.lon;

    Uri url =
Uri.parse("https://api.openweathermap.org/data/2.5/weather?lat=$lat&lon=$lon&
appid=$appID&units=metric");
    Response response = await get(url);
    data = response.body;
    if(response.statusCode == 200){
      return data;
    }else{
      return "ERROR";
    }
  }

  Future<String> GetCity(String newCity) async {
    Uri url = Uri.parse(

"https://api.openweathermap.org/data/2.5/weather?q=$newCity&appid=$appID&unit
s=metric");
    Response response = await get(url);
    data = response.body;

    if (response.statusCode == 200) {
      return data;
    } else if (response.statusCode == 400) {
      throw ErrorDescription("Cannot find data in city");
    } else if (response.statusCode == 404) {
      throw ErrorDescription("Cannot find data in city");
    } else {
      throw ErrorDescription("Cannot find data in city");
    }
  }

}
```

Weather.dart

```
class WeatherModel {
  String getWeatherIcon(int condition) {
    if (condition < 300) {
      return '🌩';
    } else if (condition < 400) {
      return '🌧';
    } else if (condition < 600) {
      return '☔';
    } else if (condition < 700) {
```

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```
      return '';
    } else if (condition < 800) {
      return '';
    } else if (condition == 800) {
      return '';
    } else if (condition <= 804) {
      return '';
    } else {
      return '';
    }
  }

  String getMessage(int temp) {
    if (temp > 25) {
      return 'It\'s  time';
    } else if (temp > 20) {
      return 'Time for shorts and ';
    } else if (temp < 10) {
      return 'You\'ll need  and ';
    } else {
      return 'Bring a  just in case';
    }
  }
}
```